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Can machine learning algorithms realistically estimate reference evapotranspiration in humid subtropical climate?

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Reference evapotranspiration (ET0) is key for irrigation water management (including amount and scheduling), especially for water-stressed regions. While empirically a robust estimation of ET0 is data intensive, machine learning (ML) algorithms can aid in ET0 estimation with ease and are evolving. Therefore, in this study six ML algorithms, particularly, Polynomial Kernel (PK), Polynomial Universal Function Kernel (PUK), Normalized Poly Kernel (NPK), Radial Basis Function (RBF), Random Forest (RF), and Locally Weighted Linear Regression (LWLR) were compared for ET0 estimation in a humid subtropical region of India. The algorithms were trained and tested against ET0 calculated by the most widely used physical model: Penman-Monteith method. The results for the algorithm training and testing indicate that at a daily time step RF yields the best ET0 estimates, followed by PUK. However, at a monthly time step, the use of multiple model evaluation matrices demonstrates that PUK performed best followed by RF. The remaining four algorithms performed unsatisfactorily at both daily and monthly time steps. These results highlight the suitability of ML algorithms in ET0 estimation and reinforce that not all ML algorithms are suitable for all analyses. Furthermore, the suitability of ML algorithms in estimating ET0 at different time steps is a-priori and is crucial.